



## 2: USING PING AND TRACEROUTE



### Theory:

How Ping and Traceroute work



### Practical:

Use Ping and Traceroute to test connectivity



### Task:

Report findings for a simulated connectivity issue

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## TASK 1



### Exact Question:

Open Command Prompt or Terminal and use the ping command to test connectivity to [www.spotify.com](https://www.spotify.com). Note the number of packets sent, received, and the average response time.



### Answer:

I opened **Command Prompt** and used the following command:

```
ping www.spotify.com
```



### Sample Findings:

| Parameter                                                                                                                     | Result |
|-------------------------------------------------------------------------------------------------------------------------------|--------|
|  Packets Sent                              | 4      |
|  Packets Received                          | 4      |
|  Packets Lost                              | 0      |
|  Packet Loss                               | 0%     |
|  Average Response Time Approximately 30 ms |        |



### Observation:

The ping was successful because all packets were received and there was **0% packet loss**.

A low average response time indicates that the destination was responding normally.

### **Conclusion:**

[www.spotify.com](https://www.spotify.com) was reachable from the system, and the connection was working successfully.

**Note:** Ping results can vary depending on the internet connection, location, DNS resolution, and whether the website responds to ICMP requests.

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## **TASK 2**

### **Exact Question:**

Use the `tracert` (tracert on Windows, `traceroute` on Linux/Mac) command to trace the path from your computer to [www.flipkart.com](https://www.flipkart.com). List the number of hops and identify the IP address of the first and last hop.

### **Answer:**

Since I am using **Windows**, I used the `tracert` command.

### **Command:**

```
tracert www.flipkart.com
```

The command displays the different **network hops** between my computer and the destination.

### **Sample Findings:**

| Finding                                                                                          | Result                                              |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|  First Hop    | 192.168.1.1                                         |
|  Total Hops   | Approximately 15                                    |
|  Last Hop     | Destination server/IP shown by <code>tracert</code> |
|  Command Used | <code>tracert www.flipkart.com</code>               |

### **Observation:**

The first hop is normally the **local router/default gateway**. The following hops represent routers or network devices used by the ISP and other networks before reaching the destination.

Some hops may display:

```
Request timed out.
```

This does **not always mean that the connection is broken**. Some routers simply do not respond to traceroute requests.

### **Conclusion:**

The `tracert` command helps identify the **path taken by network traffic** and can help locate where connectivity problems may occur.

**Note:** The exact number of hops and IP addresses can change depending on the ISP, location, routing, and current network conditions.

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## **TASK 3**

### **Exact Question:**

Simulate a connectivity issue by pinging a non-existent website (for example, ping [www.fakezomato12345.com](http://www.fakezomato12345.com)) and write down the error message you receive.

**Hint:** Compare the output to a successful ping and describe the difference in your findings.

### **Answer:**

I used the following command:

```
ping www.fakezomato12345.com
```

Because the domain name does not exist, Windows cannot resolve the hostname.

### **Expected Error:**

```
Ping request could not find host www.fakezomato12345.com.  
Please check the name and try again.
```

### **Comparison:**

#### **Successful Ping**

Hostname is resolved

Destination IP is displayed

Replies are received

Response time is shown

#### **Failed Ping**

Hostname cannot be resolved

No destination IP is available

No replies are received

Host cannot be found

### What This Indicates:

This particular error indicates a **DNS/name-resolution problem**, because the system cannot translate the domain name into an IP address.

It does not necessarily mean that the internet connection itself is down.

### Conclusion:

The test demonstrates how DNS is important for accessing websites using domain names.

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## TASK 4

### Exact Question:

Imagine you are unable to access [www.youtube.com](https://www.youtube.com) from your system. Use both **ping** and **tracert** commands to diagnose the issue and write a short report describing where the connection fails and what the possible reasons could be.

### Answer:

## Connectivity Troubleshooting Report

### Step 1: Test YouTube Using Ping

I would first run:

```
ping www.youtube.com
```

#### Possible Result:

If the ping fails, I would not immediately conclude that YouTube is down.

Some websites or networks may **block or limit ICMP ping requests**.

Therefore, I would continue with a traceroute test.

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### Step 2: Run Traceroute

On Windows, I would run:

```
tracert www.youtube.com
```

This shows the path from my computer toward the destination.

## Possible Findings:

### Case 1 – Failure at the First Hop

1 Request timed out.

Possible reasons:

-  WiFi/router problem
  -  Incorrect network configuration
  -  Local firewall
  -  Gateway problem
- 

### Case 2 – Failure After Several Hops

If the first few hops respond but later hops stop responding, the possible causes include:

-  ISP routing problem
  -  Firewall filtering
  -  Network congestion
  -  ICMP traffic being blocked
  -  Temporary routing issue
- 

### Case 3 – Traceroute Reaches the Destination but Website Does Not Open

This may indicate that the problem is not basic connectivity.

Possible reasons include:

-  DNS problem
  -  Browser problem
  -  Browser cache issue
  -  Firewall/security software
  -  Proxy configuration
  -  Website/service-specific issue
- 

## Troubleshooting Summary

| Test                                 | Purpose                     | Finding                                          |
|--------------------------------------|-----------------------------|--------------------------------------------------|
| <code>ping www.youtube.com</code>    | Check basic connectivity    | Determines whether ICMP replies are received     |
| <code>tracert www.youtube.com</code> | Check network path          | Shows where the path may stop responding         |
| <code>ipconfig /all</code>           | Check network configuration | Shows IP, gateway and DNS settings               |
| Browser Test                         | Check application access    | Determines whether the issue is browser-specific |

### 💡 Final Diagnosis:

The exact cause depends on **where the connection stops**.

If the problem occurs at the local gateway, I would check the **WiFi/router and local network configuration**.

If the problem occurs farther along the path, I would investigate a possible **ISP or routing issue**.

If the network path works but YouTube still does not open, I would check **DNS, browser, firewall, proxy, or other software settings**.

### 🏁 Final Conclusion:

Using both **Ping and Traceroute** provides more information than using only one command.

**Ping** → Checks connectivity and response

**Traceroute** → Shows the path and helps locate where a problem may occur

Therefore, these commands are useful basic tools for diagnosing network connectivity problems.

## ★ IMPORTANT COMMANDS USED

| Command                                   | Purpose                                 |
|-------------------------------------------|-----------------------------------------|
| <code>ping www.spotify.com</code>         | Test connectivity to Spotify            |
| <code>tracert www.flipkart.com</code>     | Trace the path to Flipkart              |
| <code>ping www.fakezomato12345.com</code> | Test DNS/name-resolution failure        |
| <code>ping www.youtube.com</code>         | Test YouTube connectivity               |
| <code>tracert www.youtube.com</code>      | Trace the path to YouTube               |
| <code>ipconfig /all</code>                | Check IP, gateway and DNS configuration |



# OVERALL CONCLUSION

**Ping** and **Traceroute** are two important network troubleshooting commands.

- ◆ **Ping** helps determine whether a destination responds and provides information such as packet loss and response time.
- ◆ **Traceroute (tracert on Windows)** shows the path taken by packets and can help identify where communication may be delayed or interrupted.

Using these tools together makes it easier to identify whether a problem is related to the **local network, router, ISP, DNS, routing, or destination service**.



## Simple Troubleshooting Formula:



**Ping** →



**Tracert** →



**Check IP/DNS** →



**Fix Problem** →



**Test Again**